

Yang

2100 N Central Road, Fort Lee, NJ 07024 | 217-418-6269 | yy3526@columbia.edu | [GitHub](#)

Education

Columbia University | New York, NY

Expected December 2025

Master of Science in Computer Engineering

Major courses: Cloud Computing, Natural Language Processing

University of Illinois Urbana-Champaign | Champaign, IL

August 2020-May 2024

Bachelor of Science in Computer Engineering

GPA: 3.6/4.0

Major courses: Database Systems, Algorithms, Data Structures, Communication Networks, Parallel Programming, AI

Internship Experience

Blue Vision LLC | *Software Development Intern* | Remote

August 2024-Present

- Implemented filtering using Homography and Fundamental Matrices in OpenCV, improving robustness and accuracy by **+30%**
- Integrated optimized Homography matrix computation directly into the pipeline, utilizing Eigen's Levenberg-Marquardt algorithm and confidence-weighted residuals for enhanced accuracy, resulting in a **+40%** increase in system accuracy
- Automated workflows for data handling with TinyXML, enabling efficient reading, saving, and transformation of keypoints and matches. Developed visualization tools to dynamically plot matches, streamlining debugging and analysis
- Conducted extensive testing with handmade ground truth cases, validating system performance across diverse scenarios and datasets
- Identified algorithmic limitations, such as rotation invariance, and reviewed research papers to prototype solutions, laying the groundwork for further enhancements

Huawei | *Software Development Intern* | Dongguan, China

June-August 2024

- Created Python-based tools to track and analyze data flow and latency issues in base station ping packets. The tool has features to identify source and target IPs, and to extract PDCP IDs. Engineered comprehensive features for analyzing TTI data
- Designed algorithm to calculate frame and slot differences across various datasets, reduced analysis time from 5 hours to 15 minutes
- Hands-on for over 1,000 test cases for various chip components including AALU, cCACHE, CXB, GFU, HPOE, HQM, ISA, OE, and Little core, enhancing testing protocols and efficiency using Python

Guangzhou KuGou Computer Technology Co., Ltd | *Software Development Engineer* | Guangzhou, China

July-August 2023

- Developed a server and reverse-engineered an SDK in C++ and Golang for copyright infringement detection with engineering department, detecting 10% violations
- Used Python to develop 200+ lines of scripts to generate legal drafts automatically, contributing to streamlining litigation processes and automating Legal Department workflows, reducing task completion time from 3 hours to 30 minutes
- Built a Python-based simulator designed to mimic human interactions with mobile app, enabling precise data collection
- Fostered communication between the Legal Department and R&D Development to solve technical problems

Project Experience

CampusHub - Microservices-based Event Management Platform

September 2024-December 2024

- Developed a microservices-based platform with 6 members, focusing on the Authentication Microservice while collaborating with two other microservices, an API gateway, composite layer, and frontend
- Implemented RESTful APIs for registration, login, and profile management, integrating Google OAuth for seamless user authentication and onboarding
- Developed the microservice using FastAPI, leveraging OAuth2, JWT tokens for session management via Python
- Deployed the microservice in Docker on an AWS EC2 instance, with MySQL hosted on a separate EC2 instance
- Established a CI/CD pipeline to automate deployments from GitHub and implemented asynchronous task processing

Operating System [Link](#)

November 2023-December 2023

- Developed an OS system in C with a team of 3, including file system, process execution, scheduler, and interrupt handling system
- Directed testing and debugging for key operating system features to ensure system stability and reliability

Database Development: YourFavNBAPlayer [Link](#)

September 2022-December 2022

- Developed a 5-database NBA player statistics archive on Google Cloud Platform for 217 players during the 2022 NBA playoffs
- Implementing a visual website by leveraging SQL for data management, Flask for web framework integration, and HTML for frontend design to enable advanced features such as data filtering, searching, and editing

Research Experience

Using Neural Networks to Detect Plant Phenotyping Data

Champaign, IL

Research Assistant, Supervised by Prof. Narendra Ahuja (UIUC)

July 2022-January 2024

- Automated the process of collecting and analyzing of plant growth data to support agricultural research in phenotyping studies
- Annotated 100 images with 800 leaves using VGG Image Annotator to prepare a detailed dataset for neural network training
- Developed and implemented a Fast R-CNN machine learning model in Python on a Titan GPU for analyzing plant growth with a focus on accurately measuring leaf widths for sorghum breed comparison
- Enhanced model accuracy through data augmentation and parameter optimization, leading to improved predictions
- Proposed strategic adjustments to the training dataset, resulting in a 20% improvement in model performance

Skills

- Programming Languages:** C/C++, Python, Java, Assembly, R, Golang
- Techniques, Frameworks, and Databases:** AWS, Docker, Convolutional Neural Networks, TensorFlow, PyTorch, OpenCV, Fast R-CNN, eLoFTR, MongoDB, SQL, Microsoft Office